A Brief Introduction to Programming with R

MEcon & MiQE/F Introductory Week

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Part I: Background / Tools

Roadmap

1. **Part I: Background and tools**
2. **Part II: First steps and basic concepts**
3. **Part III: Working with data**
4. **Final remarks and next steps**

Welcome

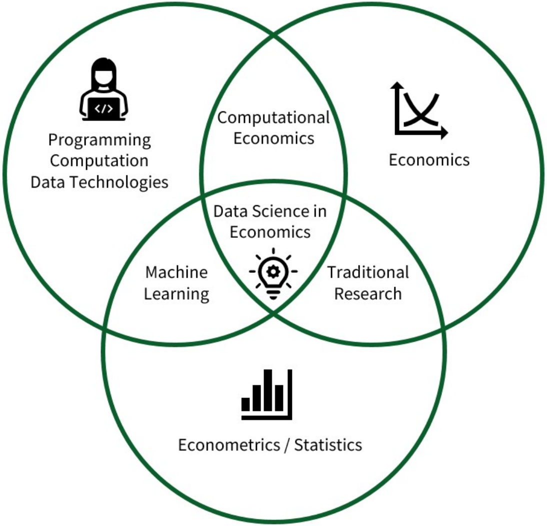
1. **Download** (or clone) the course materials
   - GitHub Repository
   - Slides, data, exercises and installation instructions are all available online
2. **Download and install R and R-studio** if you have not done so yet (https://cran.r-project.org/, https://posit.co/download/rstudio-desktop/)

Why learn to program (now)?

Background: technological change

- Data is one of the world's most **valuable resources**
- **AI and machine learning** are reshaping every business and industry

Perspective: data science and economics



Programming in the age of AI-assistance

AI assistants are powerful for many coding tasks, but they can make subtle mistakes that are difficult to spot.

- **Understand code**: Using AI effectively requires enough coding knowledge to evaluate its suggestions.
- **Problem solving**: Structuring an analysis and adapting solutions to a specific task are difficult to automate.
- **More effective prompting**: Understanding the problem helps you write precise prompts and assess the answers you receive.
- Blindly prompting AI for simple tasks hinders learning.

Recommendation: use **AI as a teacher** for programming

Why R?

A data language

- Widely used in **data science** jobs
- Less common than Python in industry, but a strong starting point: **programming fundamentals and data skills transfer across languages**
- Particularly adapted to program with **data**
- Originally designed for **statistical analysis**
- Wide and open online community: good sources to train LLM

Working directories & R projects

- **Working directory** = folder where R looks for files, e.g. data to load.
- Use `getwd()` and `setwd()`
- Better: **R Projects**
  - Define self-contained working environments
  - Save code, data, outputs together
  - Makes collaboration & reproducibility easier

Part II: First Steps and Basic Concepts

First steps in R

Variables and vectors

```r
a <- 2
a
```

```
[1] 2
```

Working with vectors

R easily allows to work with **vectors** of data!

```r
# Instantiate an integer vector 'a'
a <- c(10, 22, 33, 22, 40)
# Give names to the vector's elements
names(a) <- c("Andy", "Betty", "Claire",
              "Daniel", "Eva")
a  # Display the vector
```

```
  Andy  Betty Claire Daniel    Eva 
    10     22     33     22     40 
```

Indexing

We can access **single** (or **multiple**) elements of a vector:

```r
a[3]      # Access the 3rd element
```

```
Claire
    33
```

```r
a[3:5]    # Access the 3rd to 5th elements
```

```
Claire Daniel    Eva
    33     22     40
```

```r
a["Claire"]  # Access by name
```

```
Claire
    33
```

Inspecting variables

```r
class(a)  # Display the class
```

```
[1] "numeric"
```

```r
str(a)    # Display the structure
```

```
 Named num [1:5] 10 22 33 22 40
 - attr(*, "names")= chr [1:5] "Andy" "Betty" "Claire" "Daniel" ...
```

Basic Programming Concepts

Loops

- **Repeatedly execute** a sequence of commands
- For a **known** or **unknown** number of iterations
  - **for-loop**: number of iterations typically known
  - **while-loop**: iterate until a condition is met

For-loops in r

```r
n_iter <- 5  # Define number of iterations
# Specify the loop
for (i in 1:n_iter) {
    print(i)  # Print the number 'i'
}
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
```

For-loops: summing numbers

```r
numbers <- c(72, 42, 150, 13, 36, 19)
total_sum <- 0  # Initialize sum

# Specify the loop
for (n in numbers) {
    total_sum <- total_sum + n
}
total_sum
```

```
[1] 332
```

Nested for-loops

```
n_iter_inner <- 100
n_iter_outer <- 500

# Start outer loop
for (i in 1:n_iter_outer) {
    # Code for outer loop

    # Start inner loop
    for (j in 1:n_iter_inner) {
        # Code for inner loop
    }
}
```

How many iterations does this combination amount to?

While-loops in r

```r
# Instantiate starting value
total <- 0

# Loop while the condition is true
while (total <= 10) {
    total <- total + 3.14
}
total
```

```
[1] 12.6
```

Booleans and logical statements

```r
2 + 2 == 4  # Is 2+2 equal to 4?
```

```
[1] TRUE
```

```r
3 + 3 == 7  # Is 3+3 equal to 7?
```

```
[1] FALSE
```

```r
4 != 7      # Is 4 not equal to 7?
```

```
[1] TRUE
```

```r
!TRUE  # negation using '!'
```

```
[1] FALSE
```

```r
TRUE & FALSE # logical AND
```

Control flow with booleans

```r
condition <- TRUE

if (condition) {
    print("The condition is true!")
} else {
    print("The condition is false!")
}
```

```
[1] "The condition is true!"
```

R functions

- Functions take **parameter values** as input, process these values, and **return** results
- Many functions are **provided with R**
- Additional functions via **packages**

```r
numbers <- c(13, 25, 39, 881)
mean(numbers)     # Compute the mean
```

```
[1] 240
```

```r
sd(numbers)       # Standard deviation
```

```
[1] 428
```

```r
median(numbers)   # Median
```

```
[1] 32
```

Creating custom functions

```r
# Define custom mean function
my_mean <- function(x) {
    x_bar <- sum(x) / length(x)
    return(x_bar)
}

# Test the function
my_mean(numbers)  # call the function with the input argument
```

```
[1] 240
```

```r
mean(numbers)  # Compare with built-in
```

```
[1] 240
```

Loading functions from packages

```r
install.packages("dplyr") # Install (from CRAN): run once
library(dplyr) # Load the package: run each session

# Example: call dplyr's summarise() to compute the mean
mean_tidyverse <- data.frame(x = numbers) |>
  summarise(avg = mean(x))
as.numeric(mean_tidyverse) # display mean as number
```

```
[1] 240
```

Note: use ‘package_name::function_name()’ when multiple packages have functions of the same name to avoid ambiguities.

Data Structures and Indices

Vectors and lists

```r
# Integer vector
integer_vector <- 9:20
integer_vector[2]     # Second element
```

```
[1] 10
```

```r
integer_vector[2:5]  # Second to fifth
```

```
[1] 10 11 12 13
```

```r
# String vector
string_vector <- c("a", "b", "c")
string_vector[-3]    # All except third
```

```
[1] "a" "b"
```

Lists

Lists can contain **different types** of elements:

```r
# Create a list
my_list <- list(
    numbers = integer_vector,
    letters = string_vector,
    condition = TRUE
)
str(my_list)
```

```
List of 3
 $ numbers  : int [1:12] 9 10 11 12 13 14 15 16 17 18 ...
 $ letters  : chr [1:3] "a" "b" "c"
 $ condition: logi TRUE
```

Accessing list elements

```
# Access by name
my_list$numbers[1:3]
```

```
[1]  9 10 11
```

```
my_list[["letters"]]
```

```
[1] "a" "b" "c"
```

```
# Access by index
my_list[[1]][1:3]
```

```
[1]  9 10 11
```

Data frames: creation

```r
# Create a dataframe
my_df <- data.frame(
    Name = c("Alice", "Betty", "Claire"),
    Age = c(20, 30, 45)
)

my_df
```

Name	Age
Alice	20
Betty	30
Claire	45

Data frames: access

```r
my_df$Age              # Access column
```

```
[1] 20 30 45
```

```r
my_df[2, ]             # Second row
```

	Name	Age
2	Betty	30

Part III: Working with Data

Loading/Importing Data

Loading built-in r data sets

```
# Load built-in dataset
data(swiss)

# Check if loaded
class(swiss)
```

```
[1] "data.frame"
```

The `data()` function loads **built-in R datasets** into your environment.

Inspect the data structure

```
# Structure of the swiss dataset
str(swiss)
```

```
'data.frame':	47 obs. of  6 variables:
 $ Fertility       : num  80.2 83.1 92.5 85.8 76.9 76.1 83.8 92.4 82.
 $ Agriculture     : num  17 45.1 39.7 36.5 43.5 35.3 70.2 67.8 53.3 
 $ Examination     : int  15 6 5 12 17 9 16 14 12 16 ...
 $ Education       : int  12 9 5 7 15 7 7 8 7 13 ...
 $ Catholic        : num  9.96 84.84 93.4 33.77 5.16 ...
 $ Infant.Mortality: num  22.2 22.2 20.2 20.3 20.6 26.6 23.6 24.9 21
```

First few rows of the data

```r
head(swiss, 4)
```

	Fertility	Agriculture	Examination	Education	Catholic	Infant.Mortality
Courtelary	80.2	17.0	15	12	9.96	22.2
Delemont	83.1	45.1	6	9	84.84	22.2
Franches-Mnt	92.5	39.7	5	5	93.40	20.2
Moutier	85.8	36.5	12	7	33.77	20.3

Comma separated values (csv)

Example of **CSV format**:

```
"","Fertility","Agriculture","Examination",...
"Courtelary",80.2,17,15,...
"Delemont",83.1,45.1,6,...
```

Importing data from different sources

```r
# Csv files
swiss_csv <- read.csv("./data/swiss.csv")

# Excel files (requires readxl)
library(readxl)
swiss_excel <- read_excel("./data/swiss.xlsx")

# Spss files (requires haven)
library(haven)
swiss_spss <- read_spss("./data/swiss.sav")
```

Introduction to the Tidyverse

What is the tidyverse?

”The tidyverse is an opinionated collection of R packages designed for data science. All packages share an underlying design philosophy, grammar, and data structures.”

To install and load:

```r
install.packages("tidyverse") # Install once

library(tidyverse) # Load in each session
```

The pipe operator |>

Transform nested functions into **readable pipelines**:

```r
# Traditional approach
head(select(swiss, Fertility, Education), 3)

# With pipe operator
swiss |>
  select(Fertility, Education) |>
  head(3)
```

Note: |> is built into R (since version 4.1.0). %>% is from the `magrittr` package, which has additional features.

Select columns with `select()`

```r
# Select specific columns
swiss |>
  select(Fertility, Education, Catholic) |>
  head(3)
```

	Fertility	Education	Catholic
Courtelary	80.2	12	9.96
Delemont	83.1	9	84.84
Franches-Mnt	92.5	5	93.40

Select columns - advanced patterns

```r
# Select by pattern
swiss |>
  select(starts_with("E") | contains("Mort")) |>
  head(3)
```

	Examination	Education	Infant.Mortality
Courtelary	15	12	22.2
Delemont	6	9	22.2
Franches-Mnt	5	5	20.2

Filter rows with `filter()`

```r
# Keep rows meeting conditions
swiss |>
  filter(Education > 20) |>
  select(Fertility, Education, Catholic) |>
  head()
```

	Fertility	Education	Catholic
Lausanne	55.7	28	12.1
Neuchatel	64.4	32	16.9
V. De Geneve	35.0	53	42.3
Rive Droite	44.7	29	50.4
Rive Gauche	42.8	29	58.3

Create new variables with `mutate()`

```
swiss |>
  mutate(
    High_Education = Education > 15,
    Fert_per_100 = Fertility / 100
  ) |>
  select(Education, High_Education,
         Fertility, Fert_per_100) |>
  head(3)
```

	Education	High_Education	Fertility	Fert_per_100
Courtelary	12	FALSE	80.2	0.802
Delemont	9	FALSE	83.1	0.831
Franches-Mnt	5	FALSE	92.5	0.925

Arrange rows with `arrange()`

```
swiss |>
  arrange(desc(Education)) |>
  select(Education, Examination) |>
  head(4)
```

	Education	Examination
V. De Geneve	53	37
Neuchatel	32	35
Rive Droite	29	16
Rive Gauche	29	22

Summarize data with `summarize()`

```r
swiss |>
  summarize(
    mean_edu = mean(Education),
    sd_edu = sd(Education),
    median_fert = median(Fertility),
    n = n()
  )
```

mean_edu	sd_edu	median_fert	n
11	9.62	70.4	47

Group operations with `group_by()`

```
# Group and summarize
swiss |>
  group_by(Catholic > 50) |>
  summarize(
    mean_education = mean(Education),
    mean_fertility = mean(Fertility),
    count = n()
  )
```

Catholic > 50	mean_education	mean_fertility	count
FALSE	12.14	66.2	29
TRUE	9.11	76.5	18

Combining multiple operations

```r
swiss |>
  filter(Agriculture < 50) |>
  group_by(Catholic > 50) |>
  summarize(
    avg_exam = mean(Examination),
    n = n(),
    .groups = "drop"
  ) |>
  arrange(desc(avg_exam))
```

Catholic > 50	avg_exam	n
FALSE	23.1	15
TRUE	12.3	6

Join data frames (preparation)

```r
# Sample data
df1 <- data.frame(
  Student_ID = c(2501, 2502, 2503, 2504),
  Master = c("Mecon", "Mecon", "MiQE/F", "MiQE/F")
)

df2 <- data.frame(
  Student_ID = c(2501, 2502, 2503, 2504, 2505, 2506),
  Grades = c(6, 5, 5.5, 5.5, 4, 5)
)
```

Join data frames

```r
# Left join keeps all rows from df1
left_join(df1, df2, by = "Student_ID")
```

Student_ID	Master	Grades
2501	Mecon	6.0
2502	Mecon	5.0
2503	MiQE/F	5.5
2504	MiQE/F	5.5

Purrr: map functions

```r
# Apply a function to multiple columns
swiss |>
  select(Fertility, Agriculture, Education) |>
  map_dbl(mean) |>
  round(2)
```

```
  Fertility Agriculture   Education
       70.1        50.7        11.0
```

Visualization with R (ggplot2)

Ggplot2 basics

```
# Basic structure
ggplot(data = <DATA>,

       mapping = aes(x = <X>, y = <Y>)) +
  <GEOM_FUNCTION>() +
  <OPTIONAL_LAYERS>
```

Prepare the data

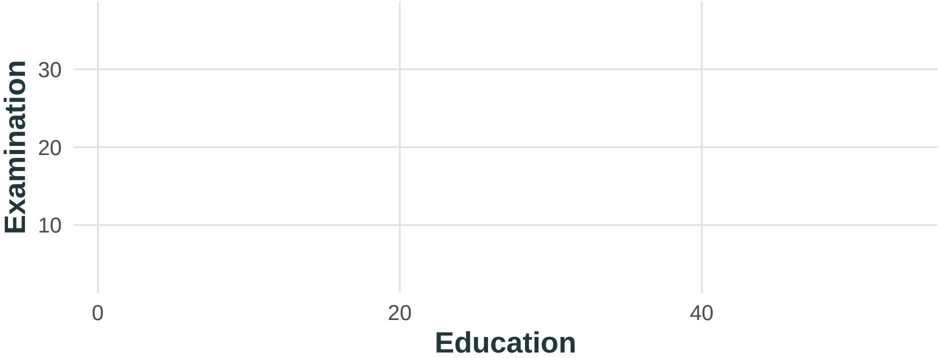
```r
# Add religion indicator
swiss <- swiss |>
  mutate(Religion = ifelse(Catholic > 50,
                          'Catholic', 'Protestant') |> factor())

# Check the data
table(swiss$Religion)
```

```
  Catholic Protestant 
        18         29 
```

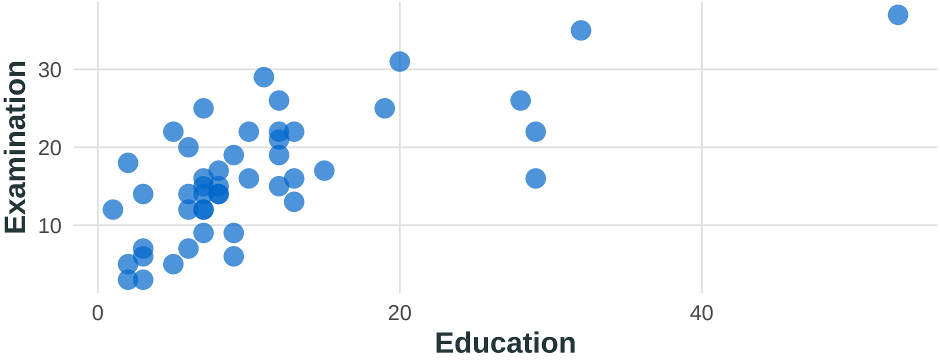
Building a plot: data + aesthetics

```
ggplot(data = swiss,
       aes(x = Education, y = Examination))
```



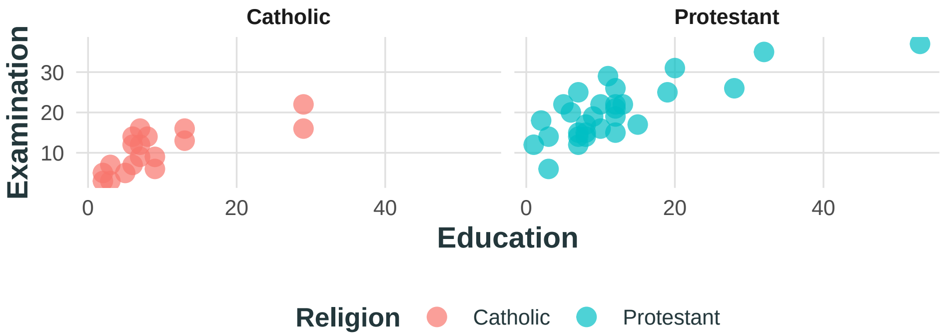
Adding geometries

```
ggplot(data = swiss,
       aes(x = Education, y = Examination)) +
  geom_point(size = 3, alpha = 0.7, color = "#0066CC")
```



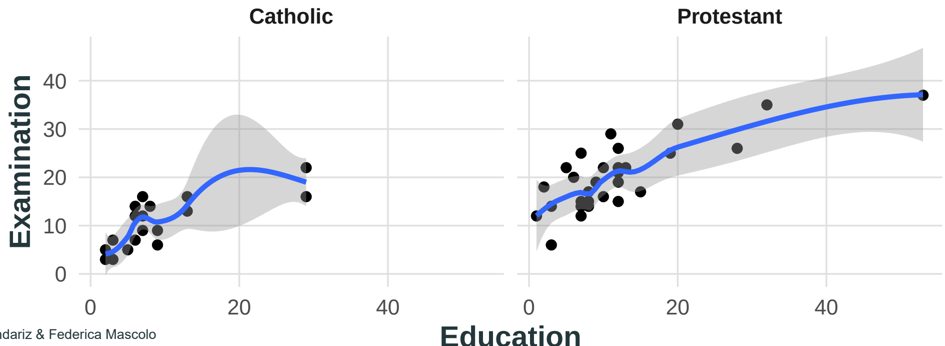
Using facets

```
ggplot(data = swiss,
       aes(x = Education, y = Examination)) +
  geom_point(size = 3, alpha = 0.7, aes(color = Religion)) +
  facet_wrap(~Religion)
```



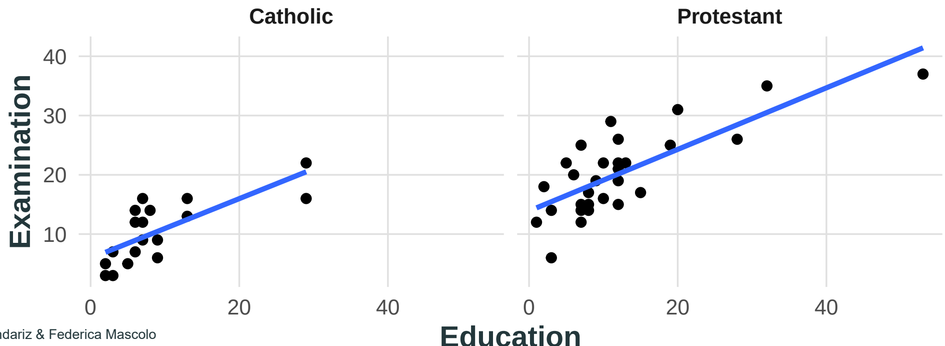
Adding statistics with loess

```
ggplot(data = swiss,
       aes(x = Education, y = Examination)) +
  geom_point() +
  geom_smooth(method = 'loess') +
  facet_wrap(~Religion)
```



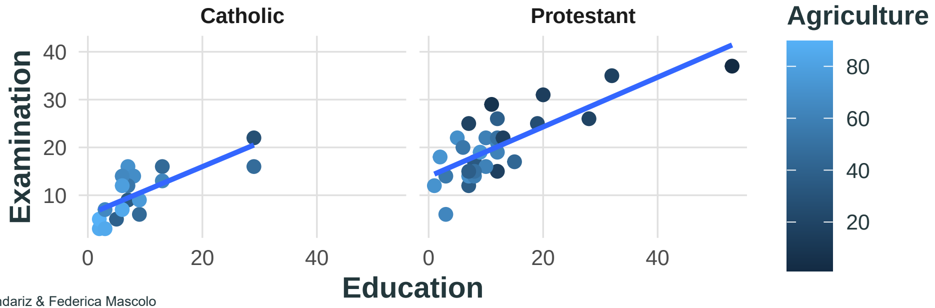
Adding statistics with lm

```r
ggplot(data = swiss,
       aes(x = Education, y = Examination)) +
  geom_point() +
  geom_smooth(method = 'lm', se = FALSE) +
  facet_wrap(~Religion)
```



Multiple aesthetics

```
ggplot(data = swiss,
       aes(x = Education, y = Examination)) +
  geom_point(aes(color = Agriculture), size = 2) +
  geom_smooth(method = 'lm', se = FALSE) +
  facet_wrap(~Religion) +
  theme(legend.position = "right")
```



Save plots

```r
# Save the last plot
ggsave("my_plot.png",
       width = 8, height = 5,
       dpi = 300)

# Save a specific plot
p <- ggplot(swiss, aes(Education, Examination)) +
  geom_point()

ggsave("scatter.pdf", plot = p,
       width = 6, height = 4)
```

Basic Statistics with R

Descriptive statistics

```r
# Create sample data
x <- c(10, 22, 33, 22, 40)

# Basic statistics
tibble(
  mean = mean(x),
  median = median(x),
  sd = sd(x),
  min = min(x),
  max = max(x)
)
```

mean	median	sd	min	max
25.4	22	11.5	10	40

T-test example

```r
# Generate sample data
set.seed(123)
sample <- rnorm(30, mean = 10, sd = 2)

# One-sample t-test
t_result <- t.test(sample, mu = 10)
t_result$p.value
```

```
[1] 0.794
```

```r
t_result$conf.int
```

```
[1]  9.17 10.64
attr(,"conf.level")
[1] 0.95
```

Linear regression - setup

```r
# Define the model formula
model1 <- Examination ~ Education

# Fit the model
fit1 <- lm(model1, data = swiss)

# View coefficients
coef(fit1) # Use 'summary(fit1)' for more details
```

```
(Intercept)   Education
     10.127       0.579
```

Multiple linear regression

```r
# Fit model with multiple predictors
fit2 <- lm(Examination ~ Education + Catholic + Agriculture,
           data = swiss)

# Model fit statistics
tibble(
  R2 = summary(fit2)$r.squared,
  Adj_R2 = summary(fit2)$adj.r.squared,
  RMSE = sigma(fit2))
```

R2	Adj_R2	RMSE
0.728	0.709	4.3

Regression coefficients table

```r
# Extract coefficient information
coef_summary <- summary(fit2)$coefficients
round(coef_summary, 3)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    18.537      2.637    7.03    0.000
Education       0.424      0.087    4.89    0.000
Catholic       -0.080      0.017   -4.75    0.000
Agriculture    -0.068      0.040   -1.71    0.095
```

Other econometric models

```
# Binary outcomes - logit
glm(y ~ x1 + x2,
    family = binomial(link = "logit"))

# Binary outcomes - probit
glm(y ~ x1 + x2,
    family = binomial(link = "probit"))

# Count data - poisson
glm(y ~ x1 + x2,
    family = poisson())

# Panel data - fixed effects
library(fixest) # install.packages("fixest")
feols(y ~ x1 + x2 | x3,
    data = panel_data)
```

Print regression results - console

```
# Basic regression table
library(modelsummary)

models <- list(
  "Model 1" = lm(Examination ~ Education,
                 data = swiss),
  "Model 2" = lm(Examination ~ Education + Catholic + Agriculture,
                  data = swiss)
)

modelsummary(models,
             stars = TRUE,
             gof_omit = "AIC|BIC|Log.Lik.")
```

Save regression output

```r
# Save table directly to file
modelsummary(models, output = "table.docx")
modelsummary(models, output = "table.html")
modelsummary(models, output = "table.tex")
modelsummary(models, output = "table.md")
modelsummary(models, output = "table.txt")
modelsummary(models, output = "table.png")

# Raw table
modelsummary(models, output = "html")
modelsummary(models, output = "latex")
modelsummary(models, output = "markdown")
```

Final Remarks

Some best practices for R programming

1. **Organization**
   - Start scripts with `library()` calls
   - Use meaningful variable names
   - Comment your code with #

2. **Efficiency**
   - Use functions to avoid repetition
   - Vectorize operations when possible
   - Use the tidyverse for data manipulation

3. **Reproducibility**
   - Set seeds for random operations
   - Use relative paths (`./data/`)
   - Work inside an R Project; document package versions

4. **Working with AI**
   - Ask for an **explanation**, not only for code
   - Give context: data, error message, what you already tried
   - **Run and read** every suggestion before you keep it
   - Never paste confidential data

Resources for learning R

Online Resources:

- Stack Overflow
- R for Data Science book
- Posit cheatsheets
- CRAN documentation

AI Assistance:

- ChatGPT
- GitHub Copilot
- Google Gemini
- Claude

Example prompts to use AI as programming teacher

Bad prompt

- *“Please solve this coding assignment.”*
- *“Here’s my code, please fix it.”*

Better prompt

"The code below yields a variable type error. Please correct it and explain my mistake."

Good prompt:

- **System prompt**
  "I am learning to code in R and want to use you as a teacher.
  Please guide me to find the solution myself, not just give the answer."
- **Question prompt**
  "This code should do [A], but instead does [B].
  I think the issue is in line [X–Y], where I meant to do [C].
  Can you give me a hint?"

Upcoming courses

Next Week

R Programming Course

Instructor: Jana Mareckova

Dates: September 7-10, 2026

Time: 9:00-12:00 and 13:00-17:00

Location: Rosenbergstrasse 30, Room 61-152

Limited spots!

Contents:

Similar material to today's workshop, taught at a slower pace, plus automated reporting with R Markdown on the final day.

Thank you!

Thanks for joining our R workshop!

Happy Coding!